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Lifting and lowering tool

Abstract

A lifting and lowering tool includes a foot configured to be supported on a surface, a bar extending therefrom, and a movable assembly including a housing, a movable platform shaped to support a load that extends from the housing, a lifting actuator configured to incrementally move the movable assembly along the bar away from the foot, a lowering actuator configured to incrementally move the movable assembly along the bar towards the foot, and a release actuator configured to disengage the movable assembly from the bar to allow free movement of the movable assembly along the bar. In embodiments, the movable assembly engages the bar through locking plates that extend through a flange of the movable assembly. In embodiments the release actuator is spaced from the lifting actuator and the lowering actuator to prevent inadvertent actuation of the release actuator. In embodiments a flange of the movable assembly surrounds the bar.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a national stage filing under 35 U.S.C. § 371 of International Application PCT/US2022/047468 filed on Oct. 21, 2022, which claims priority to U.S. Provisional Patent Application Ser. No. 63/271,632, filed on Oct. 25, 2021, each of which is incorporated herein by reference in their entireties.

FIELD OF THE INVENTION

(1) The present invention relates generally to hand tools, and more particularly to spreaders or jacks.

BACKGROUND OF THE INVENTION

(2) Conventional lifting and lowering tools, such as jacks, are known. Among other things, the present application relates to various improvements to lifting and lowering tools, which may be used for controlled raising or lowering of objects such as doors, windows, framing segments, pallets, and so on. This disclosure includes various improvements which may be utilized together or independently in various embodiments.

SUMMARY OF THE INVENTION

(3) According to an embodiment, a lifting and lowering tool includes a foot configured to be supported on a surface, a bar extending from the foot, and a movable assembly. The movable assembly includes a housing, a movable platform shaped to support a load thereon, the movable platform extending from the housing and coupled to the housing by a flange, a lifting actuator

configured to incrementally move the movable assembly along the bar away from the foot, a lowering actuator configured to incrementally move the movable assembly along the bar towards the foot, and a release actuator configured to disengage the movable assembly from the bar to allow free movement of the movable assembly along the bar. The movable assembly engages the bar through locking plates that are selectively disengaged from the bar through actuation of the lowering actuator, and the locking plates extend through the flange.

(4) According to another embodiment, a lifting and lowering tool includes a foot configured to be supported on a surface, a bar extending from the foot, and a movable assembly. The movable assembly includes a housing, a movable platform shaped to support a load thereon, a lifting actuator configured to incrementally move the movable assembly along the bar away from the foot, a lowering actuator configured to incrementally move the movable assembly along the bar towards the foot, and a release actuator configured to disengage the movable assembly from the bar to allow free movement of the movable assembly along the bar. The release actuator is spaced from the lifting actuator and the lowering actuator to prevent inadvertent actuation of the release actuator.

(5) According to another embodiment, a lifting and lowering tool includes a foot configured to be supported on a surface, a bar extending from the foot, and a movable assembly. The movable assembly includes a housing, a movable platform shaped to support a load thereon, the movable platform extending from the housing and including a flange extending from the movable platform and surrounding the bar, a lifting actuator configured to incrementally move the movable assembly along the bar away from the foot, a lowering actuator configured to incrementally move the movable assembly along the bar towards the foot, and a release actuator configured to disengage the movable assembly from the bar to allow free movement of the movable assembly along the bar.

(6) According to another embodiment, a lifting and lowering tool includes a foot configured to be supported on a surface, a bar extending from the foot, and a movable assembly. The movable assembly includes a housing, a movable platform shaped to support a load thereon, the movable platform extending above the bar away from the foot when the movable assembly is adjacent to the foot, a lifting actuator configured to incrementally move the movable assembly along the bar away from the foot, a lowering actuator configured to incrementally move the movable assembly along the bar towards the foot, and a release actuator configured to disengage the movable assembly from the bar to allow free movement of the movable assembly along the bar.

(7) These and other objects, features, and characteristics of the present invention, as well as the methods of operation and functions of the related elements of structure and the combination of parts and economies of manufacture, will become more apparent upon consideration of the following description and the appended claims with reference to the accompanying drawings, all of which form a part of this specification, wherein like reference numerals designate corresponding parts in the various figures. In one embodiment of the invention, the structural components illustrated herein are drawn to scale. It is to be expressly understood, however, that the drawings are for the purpose of illustration and description only and are not intended as a definition of the limits of the invention. In addition, it should be appreciated that structural features shown or described in any one embodiment herein can be used in other embodiments as well. As used in the specification and in the claims, the singular form of “a”, “an”, and “the” include plural referents unless the context clearly dictates otherwise.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Features of lifting and lowering tools in accordance with one or more embodiments are shown in the drawings, in which like reference numerals designate like elements. The drawings form part of this original disclosure in which:

(2) FIG. 1 illustrates a side perspective view of a lifting and lowering tool according to a first embodiment;

(3) FIG. 2 illustrates a bottom perspective view of the tool of FIG. 1;

(4) FIG. 3 illustrates a side perspective view of the tool of FIG. 1 with a housing thereof removed;

(5) FIG. 4 illustrates an isolated perspective view of a lowering and release subassembly of the tool of FIG. 1;

(6) FIG. 5A-FIG. 5C illustrate a side cross sectional view of the subassembly of FIG. 4, in neutral, incremental lowering, and released positions respectively;

(7) FIG. 6 illustrates a side perspective view of a lifting and lowering tool according to a second embodiment;

(8) FIG. 7 illustrates a side perspective view of the tool of FIG. 6 with a housing thereof removed, focusing on a movable platform thereof;

(9) FIG. 8A-FIG. 8B illustrate a side perspective view of the tool of FIG. 6 with a housing thereof removed, focusing on lowering and release actuators thereof, in incremental lowering and released positions respectively;

(10) FIG. 9 illustrates a side view of a lifting and lowering tool according to a third embodiment;

(11) FIG. 10 illustrates a side perspective view of the lifting and lowering tool of FIG. 9;

(12) FIG. 11 illustrates a cross sectional view of the embodiment of FIG. 9, showing the internal mechanism when the tool is at rest, and how pushing down on an actuator arm thereof would actuate lifting;

(13) FIG. 12 illustrates a cross sectional view of the embodiment of FIG. 9, showing the internal mechanism when the tool is at actuated by lifting on the actuator arm to actuate lowering;

(14) FIG. 13 illustrates a side view of the embodiment of FIG. 9 with a side of the housing removed, showing the internal mechanism when the tool is actuated by pressing on a release button to actuate release of a load carried by the tool.

(15) FIG. 14 illustrates a side perspective view of a lifting and lowering tool according to a fourth embodiment;

(16) FIG. 15 illustrates a side perspective view of the lifting and lowering tool of FIG. 14, with a cover of a housing thereof removed to view an interior;

(17) FIG. 16 illustrates a side view of the lifting and lowering tool of FIG. 14, with a cover of a housing thereof removed to view the interior;

(18) FIG. 17A-FIG. 17C illustrate a side cross sectional view of a subassembly of the lifting and lowering tool of FIG. 14, showing the incremental lowering and release actuators in neutral, incremental lowering, and released positions respectively;

(19) FIG. 18 illustrates an interior side view of a fifth embodiment of the lifting and lowering tool in an unactuated position;

(20) FIG. 19 illustrates the interior side view of the embodiment of the lifting and

(21) lowering tool of FIG. 18 in an actuated position;

(22) FIG. 20 illustrates a front or rear sectioned view of a sixth embodiment of the lifting and lowering tool in an unactuated position;

(23) FIG. 21 illustrates the front or rear sectioned view of the embodiment of the lifting and lowering tool of FIG. 20 in an actuated position;

(24) FIG. 22 illustrates a side sectioned view of a seventh embodiment of the lifting and lowering tool in an unactuated position;

(25) FIG. 23 illustrates the side sectioned view of the embodiment of the lifting and lowering tool of FIG. 22 in an actuated position;

(26) FIG. 24 illustrates an interior side view of an eighth embodiment of the lifting and lowering tool in an unactuated position;

(27) FIG. 25 illustrates the interior side view of the embodiment of the lifting and lowering tool of FIG. 24 in an actuated position;

(28) FIG. 26 illustrates a perspective side view of a first embodiment of a coupled assembly of an embodiment of a lifting and lowering tool and a spreader or clamp; and
(29) FIG. 27 illustrates a perspective side view of a second embodiment of a coupled assembly of an embodiment of a lifting and lowering tool and a spreader or clamp.

DETAILED DESCRIPTION OF THE ILLUSTRATED EMBODIMENT(S)

(30) FIGS. 1-5C illustrate a lifting and lowering tool **100** according to a first embodiment.

(31) As shown in FIG. 1, the tool may include a foot **110** having a front foot **110a** and a rear foot **110b** that are fixedly mounted to a bar **120**. A movable assembly **130** may be movably mounted to the bar **120**, and includes a movable platform **140** that is received between toes **150** of the front foot **110a**. It may be appreciated that the front foot **110a** and the movable platform **140** together may be placed under an object to be raised, and by actuating the movable assembly **130** the movable platform **140** may raise the object above the foot **110**. Similarly, an object placed on an elevated movable platform **150** may be lowered by actuating the movable assembly **130** to lower the object to a desired location. Details of the foot **110** are discussed in greater detail below, but as shown in FIG. 1, the rear foot **110b** may extend rearwards from the front foot **110a** and may be sized to support the tool **100** and to stop the tool **100** from toppling backwards when the tool **100** is stood without any load, especially when the movable assembly **130** is raised from the foot **110** (e.g., being fully extended therefrom).

(32) As further shown in FIG. 1, the movable assembly **130** includes a housing **160** and may include a lifting actuator **170**, such as a lever, which may be actuated against a grip **180**, a lowering actuator **190**, such as another lever, and a release actuator **200**, which could be a third lever, as further described herein. It may be appreciated that in some embodiments the lowering actuator **190** and the release actuator **200** may be ergonomically located. For example, the lowering actuator **190** may be actuatable by an extended fingertip from a hand grasping the grip **180** (less force being required to lower a raised load than to raise it) while the raising actuator **170** may be actuated by a plurality of curled fingertips as a user's hand extends from the grip **180**. As shown, the release actuator **200** may have a low mechanical advantage, and may be positioned to reduce the possibility of accidental release when the movable assembly **130** is loaded by a raised object. In a non-limiting embodiment, operating the lowering actuator **190** lowers the movable platform approximately 3 mm.

(33) While the embodiment of FIG. 1 shows the lowering actuator **190** and release actuator **200** positioned so that they are easily operated with a finger whilst the grip **180** is gripped in the hand, in other embodiments the lowering actuator **190** and release actuator **200** could be positioned at the top of the housing **160** so as to be operated by a thumb whilst the grip **180** is gripped in the hand.

(34) As shown in FIG. 1, in an embodiment, the foot **110** may be formed from a base plate **210** that is fixed to the bar **120**. In an embodiment, the foot **110** may be defined as beginning on the bar **120** where movement of the movable assembly **140** down the bar **120** is limited (e.g., by a stopper feature). As such, in an embodiment, a portion of the bar **120** may form a portion of the foot.

(35) In the embodiment of FIG. 1, a fastener **220** may extend through the bar **120** and a flange **230** extending from and fixed to the base plate **210** to secure the bar **120** to the base plate **210** of the foot **110**. In some embodiments, the base plate **210** may be arching or cranked upwards towards the extension of the bar **120**. As shown in the bottom perspective view of FIG. 2, such an angled configuration may provide clearance for a connection point **240** such as a welded connection or other fastener between the bar **120** and foot/base plate **210** as the bar **120** extends through the base plate **210**. Such a configuration may remove the need for extra finishing operations to make the weld or coupling flush with a bottom surface of the base plate **210**.

(36) In some embodiments, the foot **110** may form a triangular configuration, where the toes **150** of the front foot **110a** are splayed and meet a common rear foot **110b**, such as is described and illustrated with reference to embodiments below. It may be appreciated that a having at least three points of surface contact (e.g., a triangular shape of the foot **110**), gives a large footprint for

improved stability on uneven surfaces. As shown in FIG. 2, in an embodiment the foot **110** may form a general H or X configuration with the toes **150** of the front foot **110a** and toes **250** of the rear foot **110b** together provide four points of surface contact for the foot **110**. Similarly, in some embodiments the movable assembly **140** may have a generally triangular configuration where a portion away from the bar **120** is wider than a portion closer to the bar **120**.

(37) FIGS. 3-5C illustrate the internals of the movable assembly **130** (e.g., with the housing **160** omitted), according to an embodiment. Movement of the movable assembly **130** along the bar **120** may be understood with reference to these Figures.

(38) As shown in FIG. 3, the lifting actuator **170** pivots on a lifting actuator cross pin **260** that is attached to the housing **160** of the movable assembly **130**. Operation of the lifting actuator **170** presses on lifting plates **270**, which grip onto the bar **120**, and move the housing **160** upwards. Locking plates **280** grip onto the bar **120** preventing the housing **160** from moving back down the bar **120**. In the illustrated embodiment, as seen more clearly in FIG. 4, ends **280a** of the locking plates **280** extend into an aperture **140a** formed on an extension **140b** of the movable platform **140** that is fixed to the housing **160**, and as such, the movable platform **140** and the housing **160** maintain pressure on the locking plates **280**.

(39) It may be appreciated that as shown in FIG. 3, in some embodiments the flange **140b** of the movable platform may include apertures **140c** configured to receive fasteners therein to secure the movable platform **140** to the housing **160**. Other connections may alternatively be possible, but regardless it may be understood that the flange **140b** may be detachably coupled to housing **160** such that the movable platform **140** is detachable from the housing **160** in various embodiments. Such detachment may be advantageous in rotating the movable platform **140** to elevate the movable platform **140** above the foot **110** in some embodiments, or so that different configurations of movable platform **140** may be alternatively installed on the housing **160**, for example.

(40) Further seen in FIG. 3, as well as in FIG. 2, is that in some embodiments the foot **110** may include apertures **110c** to facilitate securing the foot **110** to a support surface. Similarly, the movable platform **140** may include apertures **140d** to facilitate securing the movable platform **140** to an object to be raised or lowered. As shown, in some embodiments, the apertures **110c** and apertures **140d** may be countersunk appropriately, such that a flat head fastener (e.g., as opposed to pan head fastener) may be received in each facing the appropriate direction to minimize protrusion and maintaining a flat configuration for the contacting surfaces of the foot **110** and the movable platform **140**.

(41) A lowering actuator cam **290**, seen more clearly in FIGS. 4 and 5, is actuated through movement of the lowering actuator **190**, and operates the locking plates **280** for controlled lowering (in the illustrated embodiment may be in approximately 3 mm increments). The lowering actuator cam **290** and the locking plates **280** may be spring biased against the housing **160** by a spring **300**, and the lowering actuator **190** and the lowering actuator cam **290** may be pivotal on the housing at a lowering actuator pivot **310**. As further shown in FIG. 3, and described in greater detail below, the release actuator **200** may pivot or actuate a release actuator cam **320**, which operates both a holding plate **330** and the locking plates **280**, disengaging the holding plate **330** and the locking plates **280** from the bar **120** and allowing free movement of the movable assembly **130** along the bar **120**, which would fully release a load being supported on the movable platform **140**. As shown, the release actuator **200** may be spring biased (bearing against the housing **160**) by release actuator spring **335**, such that release of the release actuator **200** automatically moves into an un-released position. It may be appreciated that a spring **340** between the holding plate **330** and the lifting plates **270** biases the relative relationship between the lifting plates **270** and the holding plate **330** to permit walking the movable assembly **130** along the bar **120** and holding in an elevated position. Similarly, the spring **300** between the locking plates **280** and the housing **160** prevent unintended lowering of the load until incrementally moved by the lowering actuator **190**, or release by the release actuator **200**.

(42) As shown in FIG. 4, depicting the holding plate **330**, the locking plates **280**, the release actuator **200**, and the lowering actuator **190** in isolation, operating the release actuator **200** disengages the holding plate **330** and the locking plates **280** from the bar **120**, allowing free movement of the movable assembly **130** up or down the bar **120** (omitted from the isolated subassembly depicted, but understood as passing through the apertures of the holding plate **330** and locking plates **280**). As shown, the release actuator cam **320** may be received in a gap **340** between a pair of lowering actuator cams **290** in some embodiments, so that the release actuator cam **320** may act directly on the holding plates **280**.

(43) The relative operations of the lowering actuator **190**, the release actuator **200**, and the locking plates **280** and the holding plate **330** may be better understood with reference to FIGS. 5A-5C, depicting cross sectional views bisecting the subassembly shown in isolation in FIG. 4. FIG. 5A illustrates the subassembly at rest, as holding onto the bar **120**. As shown in FIG. 5B, pulling on or otherwise rotating the lowering actuator **190** causes the lowering actuator cams **290** to press on the locking plates **280** and pulling the bar **120** upward, moving the movable assembly **130** down towards the foot **110**. As the holding plate **330** remains engaged on the bar **120**, movement is limited to the aforementioned increment. As shown in FIG. 5C, however, actuating or in the illustrated embodiment the rotating the release actuator **200** in a counterclockwise direction (e.g., lifting it away from the lowering actuator **190**) pushes down on the release actuator cam **320**, which pushes against the holding plates **280**, and by virtue of the engagement between the holding plate **330**, the release lever cam **320**, and an aperture **350** in the release actuator cam **320** that allows a pivot **360** for the lowering actuator **190** to not be impacted by the movement of the release actuator cam **320**, actuating the release actuator **200** causes disengagement of both the holding plate **330** and the lowering plates **280**, such that the movable assembly **130** freely moves along the bar **120** without being limited to the incremental movement.

(44) As illustrated in FIGS. 6-8, in another embodiment a lifting and lowering tool **500** may be generally similar to other lifting and lowering tools disclosed herein, except as noted. As shown, the lifting and lowering tool **500** may include a foot **510** that include features similar to those described as to foot **110**, such as a front foot **510a** similar to front foot **110a**. As shown in the illustrated embodiment, however, in some embodiments rear foot **510b** may be being unitary while front foot **510a** is splayed, such that front foot **510a** and rear foot **510b** form a tripod forming three points of contact on a support surface. As further shown, extending from the foot **510** is bar **520**, which may be generally similar to bar **120**. Movable assembly **530** may be configured to move along the bar **520**, and support a movable platform **540** that when in a lowered position is received between toes **550** of the front foot **510a**, and when raised alongside a housing **560** of the movable assembly **530** may lift a load. As described in greater detail below, raising the movable assembly **530** may be accomplished through actuating a raising actuator **570**, which may be pulled against a grip **580** formed with the housing **560** or extending from the housing **560**. Similarly, lowering the movable assembly **530** may be accomplished by actuating lowering actuator **590** which may incrementally lower the movable assembly. Fully releasing the movable assembly **530** may be accomplished by actuating a release actuator **600**.

(45) Turning to FIG. 7, depicting portions of the movable assembly **530** with the housing **560** removed, it may be appreciated that in some embodiments, the movable platform **540** is separately carried on the bar **520**, and would be coupled to the locking plates **610** (described in greater detail below) by the housing **560**. As shown, such a movable platform **540** may extend from a movable platform flange **620** which may envelop the bar **520**. For example, a generally rectangular hole **630** may be formed in the movable platform flange **620**. It may be appreciated that should the connection between the housing **560** and the movable platform **540** fail in operation, this hole **630** and the angled connection between the bar **520** and the hole **630** allows the movable platform **540** to lock onto the bar **520**, preventing the load from lowering suddenly. Accordingly, in some embodiments, the flange **620** may act as a locking plate **610** if the movable platform **540** were to

separate from the housing **560** (e.g., from failure of the housing **560** or fasteners coupling the movable platform **630** and/or the flange **620** to the housing **560** (e.g., supporting the movable platform **630** and/or the flange **620** to move with the movable assembly).

(46) While the movable assembly **530** may include a lifting actuator generally similar to the lifting actuator **170** described above, it may be appreciated that in some embodiments the mechanism of a lowering actuator or release actuator may differ from that of lowering actuator **190** and release actuator **200**. For example, as illustrated in FIGS. **8A** and **8B**, where the movable platform **540** is separately carried on the bar **520** and coupled to the locking plates **610** by the housing **560** such as in the case of the lifting and lowering tool **500**, a locking plates cross pin **640** or other bearing feature may be connected to the housing **560** and maintain pressure on the locking plates **610**. As further shown, a pin and spring connection **650** between a holding plate **660** and the locking plates **610** may be provided to appropriately bias the locking plates **610** relative to the holding plate **660**, and provide a flexible connection between the holding plate **660** and the locking plates **610**. As such, in embodiments including the pin and spring connection **650** between the holding plate **660** and the locking plates **610**, the lowering actuator **590** actuates by disengaging the locking plates **610** from the bar **520** while maintaining the holding plate **660** action on the bar **520**. As noted in above, operating the lowering actuator **590** may lower the movable assembly **530** approximately 3 mm. This amount is determined by the lowering actuator cam **670** coupled to the lowering actuator **590** pressing on the locking plates **610** to walk the movable assembly towards the foot **510**. It may be appreciated that the approximately 3 mm movement of the illustrated embodiment is based on the relative size of the actuator arm, plates and the bar, and that other incremental lowering amounts may be utilized by increasing or decreasing the throw of the cam on the lowering actuator. Similarly, the release actuator **600** is shaped to, when actuated, press on the holding plate **660** and the locking plates **610**, so as to disengage them from the bar **520** and allow free movement of the movable assembly **530** along the bar **520**.

(47) FIGS. **9-13** show another embodiment of the present disclosure, configured to use a common lifting and lowering actuator.

(48) As shown in FIG. **9**, a lifting and lowering tool **1000** may include a trigger actuator **1010** that may be pushed down from a neutral position to raise a movable platform **1020** of a movable assembly **1030** along a bar **1040**, or may be raised up from a neutral position for incremental lowering of the movable assembly **1040** (and thus the movable platform **1020**) down the bar **1040**. Similar to the movable assemblies of other embodiments described herein, the movable assembly **1030** may include a housing **1050** from which the movable platform **1020** is coupled to, and from which the trigger actuator **1010** extends. In some embodiments, It may be appreciated that a stationary grip extending from the housing **1050** and configured to be engaged when pushing the trigger actuator **1010** down towards it may provide a more stable feel for the user, and offer the option of squeezing the trigger down versus simply pushing the trigger down, regardless of if such stationary grip is not utilized when lifting the trigger actuator **1010** up to incrementally lower the movable platform **1020**.

(49) As further described herein, a separate release actuator **1060** may be provided and required to be pressed to fully release engagement of the movable assembly **1030** from the bar **1040**. As shown, the full release actuator **1060** may be spaced from and separated from the trigger actuator **1010** so as to prevent inadvertent full release, which could drop a load supported by the movable platform **1020**. While in some embodiments the lifting and lowering tool **1000** may include a foot similar to the foot **110**, in an embodiment, the foot may differ as shown. As seen in FIG. **9**, a foot **1070** may include a front foot **1070a** and a rear foot **1070b**, where the front foot **1070a** may be removably attached to the rear foot **1070b** to allow for replacement or attachment of alternate configuration of the base plate to reshape the foot.

(50) As seen in the perspective view of the lifting and lowering tool **1000** of FIG. **10**, in some embodiments one or more of the movable assembly **1030** and the foot **1070** may have holes **1080**

to allow the lifting and lowering tool **1000** to be rigidly attached to another object. In an embodiment, holes **1080** in the movable assembly **1030** or the movable platform **1020** thereof may be added to allow users to attach extensions or other custom-made additions to suit their particular needs. In an embodiment, the handle on the trigger actuator **1010** could be extendable to allow for additional leverage when raising the movable assembly. For example, an outer handle **1010a** may be slidable relative to an inner handle **1010b**, interlocking but selectively releasable via a handle actuator **1010c**.

(51) An embodiment of the movable assembly **1030** is depicted in FIG. **11** as a cross sectional view inside the housing **1050**. As shown, the trigger actuator **1010** and release actuator **1060** as received within the housing **1050** are depicted. As shown, the engagement of the trigger actuator **1010** as alternatively actuating both lifting and lowering of the movable assembly **1030** may be appreciated through engagement between the trigger actuator **1010** and an internal lowering actuator **1090**, and between the trigger actuator **1010** and the lifting plates **1100**, or the internal lowering actuator **1090** and the locking plates **1110** and/or holding plate **1120**.

(52) As shown in FIG. **12**, when the trigger **1010** is raised to actuate lowering the movable assembly **1030**, a lowering mechanism linkage **1130** between the trigger actuator **1010** and lowering actuator **1090** is also raised. This causes the lowering actuator **1090** to rotate, and a cam/contact point **1140** presses down on the locking plates **1110** and slightly raising the holding plate **1120**. This allows the locking plates **1110** to be pushed down the bar **1040**, incrementally lowering the movable assembly **1030**, while the holding plate **1120** is raised to grip the bar **1120** and prevent full release of the load.

(53) FIG. **13** shows an embodiment of actuation of the release actuator **1060** of the lifting and lowering tool **1000**. As shown, pressing down on the release actuator **1060** disengages the holding plate **1120** and the locking plates **1110** from the bar **1040**, allowing free movement of the movable assembly **1050** along the bar **1040**. It would be understood that while under load, or while unsupported by a user, gravity would pull the movable assembly **1030** down the bar **1040**.

(54) FIGS. **14-17C** illustrate another embodiment of a lifting and lowering tool, namely lifting and lowering tool **1500**. As shown in FIG. **14**, the lifting and lowering tool **1500** includes a foot **1510** that is fixedly mounted to a bar **1520**. It may be appreciated that the foot **1510** and bar **1520** may be generally the same as the bar **110** and bar **120** described herein, or may be similar to the foot **510** and bar **520**, or may be similar to the foot **1070** and bar **1040**, or combinations thereof. A movable assembly **1530** may be movably mounted to the bar **1520**, and includes a movable platform **1540** that may be received between toes **1550** of the foot **1510**.

(55) FIG. **15** shows another perspective view of the tool **1500**, with a housing **1560** of the movable assembly **1530** having a cover portion **1560a** (as seen in FIG. **14**) removed to show the interior thereof. As shown, a lifting actuator **1570**, such as a lever, may extend from the housing **1560**, and may be actuated against a grip **1580** fixed or formed with the housing **1560**. In the illustrated embodiment, the lifting actuator **1570** is configured to be pushed in a downward direction to lift the movable platform **1540**. It may be appreciated that such a configuration may permit a user to leverage their body weight, either applied through a hand or even a foot, to assist in lifting a load on the movable platform **1540**. As further shown in FIG. **15**, the tool **1500** includes a lowering actuator **1590**, such as another lever, and a release actuator **1600**, which could be an actuator button as shown, or could be another lever such as may be appreciated with reference to the other embodiments described herein.

(56) It may be appreciated that the lifting actuator **1570** acts on lifting plates **1670**, incrementally raising the movable platform **1540** along the bar **1520**, similar to other embodiments described herein, albeit with the lifting plates **1670** angled and beating against the housing **1560** so that downward action of the lifting actuator **1570** towards the foot **1510** raises the movable assembly **1530** away from the foot **1510**. A spring **1675** bears between the lifting plates **1670** and the housing **1560** to act as a return bias for the lifting lever **1570** and facilitate the incremental walking action

for the lifting plates **1670**.

(57) As described in greater detail below, the lowering actuator **1590** and the release actuator **1600** may engage with locking plates **1680**. As shown, the locking plates **1680** include ends **1680a** that extend into an aperture **1540a** formed on an extension **1540b** of the movable platform **1540** that is fixed to the housing **1560**. This may be understood as similar to the configuration of the locking plates **280** in relation to the movable platform **140** of the tool **100**, albeit opposite in position as engaging the bar **1520** as illustrated. As such, it may be appreciated that the movable platform **1540** and the housing **1560** maintain pressure on the locking plates **1680** with such a connection.

(58) Features of the lowering actuator **1590**, the release actuator **1600**, and the operation thereof may be understood with reference to FIG. **16** and FIGS. **17A-C**. Specifically, as shown in FIG. **16**, a lowering actuator cam **1690** is actuated through movement of the lowering actuator **1590**, and presses on the locking plates **1680**, moving the locking plates **1680** and the bar **1520** relative to a holding plate **1730** for controlled lowering of the movable assembly **1530** towards the foot **1510** (in the illustrated embodiment may be in approximately 3 mm increments). The locking plates **1680** may be spring biased against a portion of housing **1560** by a spring **1700**. In an embodiment this may in turn spring bias the lowering actuator **1590**, while in other embodiments, such as that illustrated, a torsion spring (e.g., torsion spring **1760** described below) on the lowering actuator **1590** may bias the lowering actuator **1590** into an unactuated position. The spring **1700** between the locking plates **1680** and the housing **1560** prevent unintended lowering of the load until incrementally moved by the lowering actuator **1590**, or release by the release actuator **1600**. While in some embodiments the lowering actuator **1590** and the lowering actuator cam **1690** may be pivotal on the housing **1560**, in the illustrated embodiment the lowering actuator **1590** is linked to the release actuator **1600** as described below, which is borne by the housing **1560**. As shown, the release actuator **1600** may actuate a release actuator cam **1720**, which causes both the holding plate **1730** and the locking plates **1680** to disengage from the bar **1520** and allowing free movement of the movable assembly **1530** along the bar **1520**, which would fully release a load being supported on the movable platform **1540**. It may be appreciated that a spring **1740** between the holding plate **1730** and the housing **1560** biases the holding plate **1730** and the locking plates **1680** to permit walking the movable assembly **1530** down the bar **1520** towards the foot **1510**, and holding the movable assembly **1530** gripping the bar **1520** in an elevated position.

(59) The engagement of the lowering actuator **1590**, the release actuator **1600**, the locking plates **1680** and the holding plate **1730** may be easier understood with reference to FIGS. **17A-17C**, depicting cross sectional views bisecting a subassembly of those components. FIG. **17A** illustrates the subassembly at rest, as it would be holding onto the bar **1520** to prevent movement of the movable assembly **1530** relative to the bar **1520**. As shown, the release actuator cam **1720** may pass through an aperture **1750** in the lowering actuator **1590**, so that the lowering actuator **1590** may be actuated without engaging the release actuator cam **1720**.

(60) As shown in FIG. **17B**, pulling on or otherwise rotating the lowering actuator **1590** causes the lowering actuator cam **1690** to press on the locking plates **1680**, pulling the bar **1520** upward, moving the movable assembly **1530** down towards the foot **1510**. As the holding plate **1730** remains engaged on the bar **1520**, movement is limited to the aforementioned increment. It may be appreciated that in an embodiment, a spring **1760** (e.g., a torsion spring in the illustrated embodiment, configured to bear against the housing **1560** and a recess in the lowering actuator **1590**) may return the lowering actuator **1590** to its unactuated position.

(61) As shown in FIG. **17C**, however, the release actuator cam **1720** may be shaped to engage with both the holding plate **1730**, and the lowering plates **1680**, such that actuating or in the illustrated embodiment the pressing the release actuator **1700** pushes down on the release actuator cam **1720**, which pushes against the holding plates **1680**, and by virtue of the engagement between the holding plate **1730**, the release lever cam **1720**, and the aperture **1750** in the release actuator cam **1590**, causes disengagement of both the holding plate **1730** and the lowering plates **1680** from the bar

1520, such that the movable assembly **1530** freely moves along the bar **1520** without being limited to the incremental movement. It may be appreciated that in an embodiment, a spring **1770** (e.g., a torsion spring in the illustrated embodiment, configured to bear against the housing **1560** and a recess in the release actuator **1600**) may return the release actuator **1600** and the release actuator cam **1720** their unactuated positions.

(62) FIG. **18** illustrates a side cross sectional view of an embodiment of a lifting and lowering tool **2000**, showing the internal mechanism inside the housing **2010**, of another embodiment in which a lowering trigger **2020**, configured to be actuated by a user's thumb, is connected to the lowering plate **2030** by a connecting link **2040** or other linkage. Pressing down on the lowering trigger **2020** lifts an inboard end of the trigger **2020** by rotating the trigger **2020** about a pivot **2050**, causing the lowering plate **2030** to rotate about a pin connection **2060** with the holding plate **2070**. As the lowering plate **2030** rotates, the locking plates **2080** are pushed downward. The incremental lowering position is shown in FIG. **19**.

(63) As shown in a front or rear cross-sectional view of an embodiment of a lifting and lowering tool **2100** in FIG. **20**, in some embodiments a pushbutton mechanism may be configured to provide full release of the lift. In such an embodiment one or more push buttons **2110** may be held in the housing **2120** and cover **2130**. In an embodiment, a pin **2140** is used to maintain alignment of the buttons **2110**. A spring **2150**, as shown, may return the buttons **2110** to their disengaged position when released. As shown, pressing the one or more buttons **2110** inward may engage the cam surfaces **2160** against a top of the lowering plate **2170**, in the area just above a pin **2180** connecting the lowering plate **2170** to the holding plate **2190**. As the cam surfaces **2160** advance, the lowering plate **2170**, holding plate **2190** and locking tabs **2200** may be pushed down until they disengage with the bar **2210**, allowing the movable assembly to be freely moved up and down the bar. The released position for the embodiment in FIG. **20** is shown in FIG. **21**.

(64) FIG. **22** illustrates an alternative mechanism for providing the incremental lowering function, as implemented in an embodiment of a lifting and lowering tool **2300**. The lowering trigger **2310** may be activated by a user pressing down on it. In the illustrated embodiment, a lowering trigger cam surface **2320** contacts the lowering lever **2330**, causing it to pivot downward about the lever's pivot point **2340**. A cam surface **2350** on the lowering lever pushes down on the locking tabs **2360**. The incremental lowering position for the embodiment in FIG. **22** is illustrated in FIG. **23**.

(65) FIG. **24** shows another side cross sectional view of an internal mechanism of an embodiment of a lifting and lowering tool **2400**, inside the housing **2410** thereof, showing an alternative mechanism for providing the incremental lowering function. As illustrated a lowering trigger **2420** is connected to the lowering plate **2430** by a connecting link **2440**. When a user pushes the lowering trigger **2420** towards the bar **2450** (e.g., by rotation about a pivot pin **2460**), the movement lifts the outboard end of the trigger **2420**, and via the connecting link **2440** lifts the outboard end of the lowering plate **2430**. This movement causes the lowering plate **2430** to rotate about a pin connection **2470** with a holding plate **2480**. As the lowering plate rotates, the locking tabs **2490** are pushed downward. The incremental lowering position for the embodiment of FIG. **24** is shown in FIG. **25**.

(66) Finally, FIG. **26** illustrates an embodiment of a coupler **2500** between a support extension **2510** (e.g., a spreader or a clamp, including as illustrated a spreader configuration for a bar clamp) and an embodiment of a lifting and lowering tool **2520** similar to or appreciable from any of the embodiments disclosed above. It may be understood that the lifting and lowering tool **2520** may be coupled with the support extension **2510** via the coupler **2500** that interconnects movement of the movable assembly with the support extension **2510**. In the embodiment of FIG. **26** the coupler **2500** may be a bar coupler configured as a sleeve that engages with a housing **2530** of the lifting and lowering tool **2520** and as such, movement of the movable assembly **2540** carrying the housing **2530** pushes the bar coupler **2500** (and the support extension **2510** coupled thereto) along the bar **2550** of the lifting and lowering tool **2520**. In an embodiment, a bar **2510a** of the support extension

2510 may be side by side with the bar 2550 of the lifting and lowering tool 2520 when the coupler 2500 is supporting the support extension 2510 on the movable assembly 2530 of the lifting and lowering tool 2520. For example, in the illustrated embodiment, the bar 2550 may be extending behind (and as viewed eclipsed by) the bar 2510a of the support extension 2510. It may be appreciated that the support extension 2510 may be considered a part of a movable platform 2570 of the movable assembly 2540, and that in various embodiments it could be construed that the movable platform 2570 or the movable assembly 2540 extends above the bar 2550 away from a foot 2555 of the lifting and lowering tool 2520 when the movable assembly 2540 is positioned adjacent to the foot 2555 (e.g., at a lowest point of lowering of the movable assembly 2540 along the bar 2550).

(67) Similarly, as shown in FIG. 27, a movable platform coupler 2560 may slide on or otherwise be fastened to the movable platform 2570 of the lifting and lowering tool 2520, and as such, movement of the movable assembly 2540 may similarly raise or lower the support extension 2510. Such configurations may be useful for lifting and lowering larger objects, such as cabinetry, which may be secured by a spreader or clamp configuration of the support extension 2510 in a manner that balances the cabinetry with respect to the lifting and lowering tool, and may secure the cabinetry via the spreading or clamping configuration of such a support extension 2510 (e.g., spreading within the interior of the cabinetry, or clamping to a surface thereof).

(68) Among the various embodiments disclosed herein, it may be appreciated that a spaced arrangement separating the lowering actuators and the release actuators thereof, and/or the ergonomic engagement of the release actuator being different from the ergonomic engagement of the lowering actuator may prevent an inadvertent use of the release actuator, which could drop a load on a movable platform resulting in damage to the load or injury to a user. For example, where the lowering actuator is a lever, the release actuator could be a pushbutton. Other arrangements may be appreciated facilitating such a configuration. For example, where a lifting actuator and/or the lowering actuator is engaged by a user's palm or index finger, the release actuator may be shaped or positioned to be actuated by the user's thumb. In some such embodiments one or more side buttons may be configured to fully release the load. In some such embodiments, full release may be accomplished by pressing the one or more buttons into the housing, which may disengage both the holding and locking plates allowing the housing assembly to be freely moved up and down the bar. In some embodiments, a pair of buttons may both need to be actuated to release the holding and locking plates, or a pair of buttons may be redundant (e.g., only one button needs to be pressed) but both are provided for either left-handed or right-handed engagement. It may be appreciated that requiring both buttons to be pressed may provide improved safety in preventing accidental full release of the load.

(69) In yet another embodiment, a safety interlock may be built into the lifting arm. When a load is present on the movable platform, the interlock will be pressed down against the movable platform. This will engage a linkage coupled to the lowering actuator which limits how far the lowering actuator can be pressed, so only an incremental lowering can be performed. Once incrementally lowered so the load is removed, the interlock and associated linkage to the lowering actuator disengages, allowing the lowering actuator to be additionally depressed into the full release mode, where the housing assembly can be freely moved up and down the bar.

(70) The objects, features, and characteristics of the present invention, as well as the methods of operation and functions of the related elements of structure and the combination of parts and economies of manufacture, may be apparent upon consideration of the description and drawings herein, all of which form a part of this specification. In one embodiment of the invention, the structural components illustrated herein are drawn to scale. It is to be expressly understood, however, that the drawings are for the purpose of illustration and description only and are not intended as a definition of the limits of the invention. In addition, it should be appreciated that structural features shown or described in any one embodiment herein can be used in other

embodiments as well. As used herein, the singular form of “a”, “an”, and “the” include plural referents unless the context clearly dictates otherwise.

(71) In various embodiments, the lifting and lowering tool described herein may be formed from metal, plastic, ceramic, wood, or any other appropriate material, or combinations of such materials. It may be appreciated that the components described herein may be of different constructions or configurations, including but not limited to one or more being comprised of different material choices. For example, various components described herein may each be constructed from a variety of materials, including but not limited to one or more of fabrics, plastics, metals, rubbers, elastomers, or any other appropriate material choice, such as aluminum (e.g., machined aluminum), iron (e.g., steel), ceramic, or any other appropriate material. In addition, portions of tools leveraging the above teachings may be formed from molded plastic, metal, or combinations thereof (e.g., plastic with metal supports or fasteners coupling portions together). In some embodiments, structural and functional components may be formed from metal or hard plastic, while exterior-most gripped components positioned to engage the palm of a gripping hand to provide the palm with a comfortable gripping surface may be made of a suitable molded plastic material or elastomeric material, and may be generally formed as a bi-material suitable molded plastic material coated with a layer of an elastomeric material, such as a rubber-based material. In some embodiments, the material choices may differ from component to component. In various embodiments, some components may be integrally formed together, while other components may be assembled by any appropriate mechanism, including but not limited to fastened, welded, snap-fit, friction fit, adhesive bonding, or other appropriate securements.

(72) Although the invention has been described in detail for the purpose of illustration based on what is currently considered to be the most practical and preferred embodiments, it is to be understood that such detail is solely for that purpose and that the invention is not limited to the disclosed embodiments, but, on the contrary, is intended to cover modifications and equivalent arrangements that are within the spirit and scope inventions more specifically understood in the context of the above disclosure.

Claims

1. A lifting and lowering tool comprising: a foot configured to be supported on a surface; a bar extending from the foot; and a movable assembly comprising: a housing; a movable platform shaped to support a load thereon; a lifting actuator configured to incrementally move the movable assembly along the bar away from the foot; a lowering actuator configured to incrementally move the movable assembly along the bar towards the foot; and a release actuator configured to disengage the movable assembly from the bar to allow free movement of the movable assembly along the bar; wherein the release actuator is spaced from the lifting actuator and the lowering actuator to prevent inadvertent actuation of the release actuator; and wherein the release actuator and the lowering actuator extend from a common face of the housing, with the release actuator above the lowering actuator relative to the foot, and the lowering actuator is further extended from the common face than the release actuator.
2. The lifting and lowering tool of claim 1, wherein the movable platform is coupled to the housing by a flange; the movable assembly engages the bar through locking plates that are selectively disengaged from the bar through actuation of the lowering actuator; and the locking plates extend through the flange.
3. The lifting and lowering tool of claim 1, wherein the movable platform extends above the bar away from the foot when the movable assembly is adjacent to the foot.
4. The lifting and lowering tool of claim 3, wherein the movable platform comprises a spreader or a clamp.
5. The lifting and lowering tool of claim 1, wherein the movable platform extends from the housing

and includes a flange extending from the movable platform and surrounding the bar.

6. The lifting and lowering tool of claim 1, wherein a portion of the housing is between the release actuator and the lowering actuator.
